

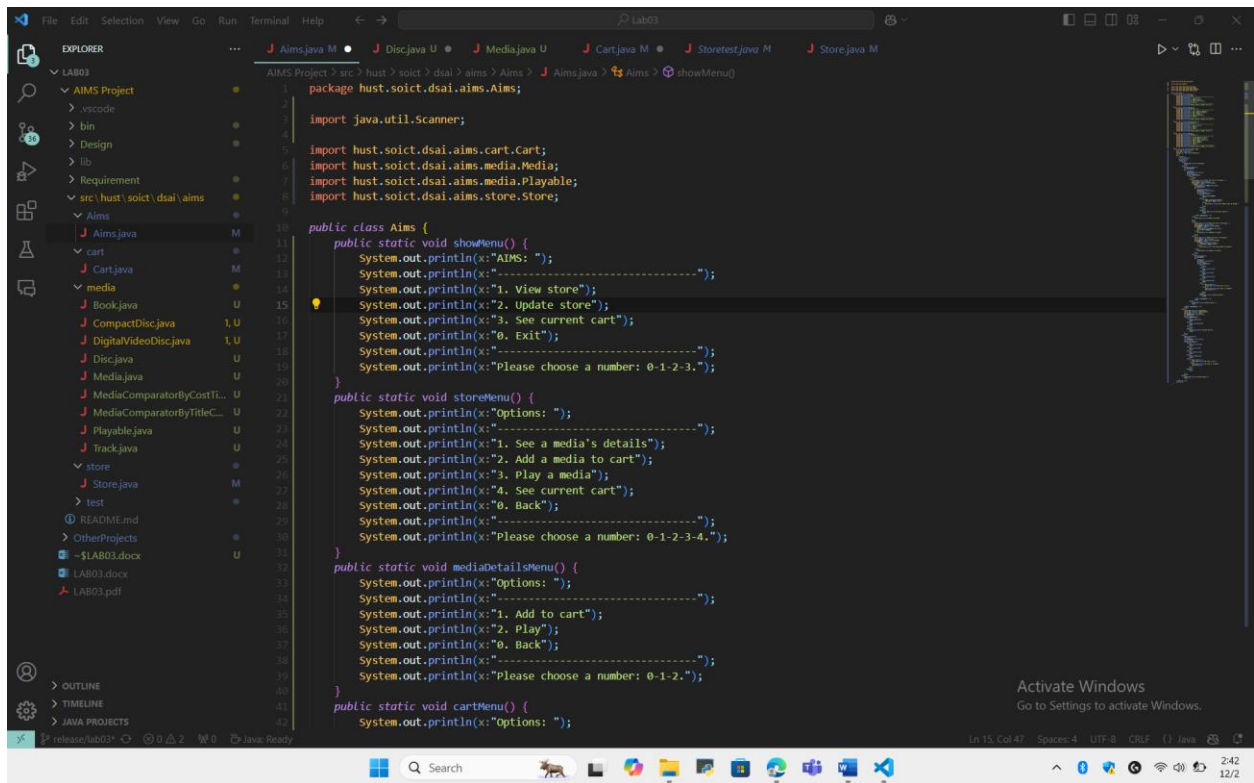
# Report Lab04

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## 1. Source code

### 1.1 Aims class



```
package hust.soict.dsai.aims.Aims;

import java.util.Scanner;

import hust.soict.dsai.aims.cart.Cart;
import hust.soict.dsai.aims.media.Media;
import hust.soict.dsai.aims.media.Playable;
import hust.soict.dsai.aims.store.Store;

public class Aims {

    public static void showMenu() {
        System.out.println(x:"AIMS: ");
        System.out.println(x:"-----");
        System.out.println(x:"1. View store");
        System.out.println(x:"2. Update store");
        System.out.println(x:"3. See current cart");
        System.out.println(x:"0. Exit");
        System.out.println(x:"-----");
        System.out.println(x:"Please choose a number: 0-1-2-3.");
    }

    public static void storeMenu() {
        System.out.println(x:"Options: ");
        System.out.println(x:"-----");
        System.out.println(x:"1. See a media's details");
        System.out.println(x:"2. Add a media to cart");
        System.out.println(x:"3. Play a media");
        System.out.println(x:"4. See current cart");
        System.out.println(x:"0. Back");
        System.out.println(x:"-----");
        System.out.println(x:"Please choose a number: 0-1-2-3-4.");
    }

    public static void mediaDetailsMenu() {
        System.out.println(x:"Options: ");
        System.out.println(x:"-----");
        System.out.println(x:"1. Add to cart");
        System.out.println(x:"2. Play");
        System.out.println(x:"0. Back");
        System.out.println(x:"-----");
        System.out.println(x:"Please choose a number: 0-1-2.");
    }

    public static void carMenu() {
        System.out.println(x:"Options: ");
    }
}
```

```
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LAB03
  AIMS Project
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  Aims
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Aims.java
10 public class Aims {
11     public static void mediaDetailsMenu() {
12     }
13     public static void cartMenu() {
14         System.out.println(x:"Options: ");
15         System.out.println(x:"1. Filter medias in cart");
16         System.out.println(x:"2. Sort medias in cart");
17         System.out.println(x:"3. Remove media from cart");
18         System.out.println(x:"4. Play a media");
19         System.out.println(x:"5. Place order");
20         System.out.println(x:"0. Back");
21         System.out.println(x:"-----");
22         System.out.println(x:"Please choose a number: 0-1-2-3-4-5.");
23     }
24 }
Run | Debug
public static void main(String[] args) {
    Store store = new Store();
    Cart cart = new Cart();
    Scanner sc = new Scanner(System.in);
    int a;
    do {
        showMenu();
        a = sc.nextInt();
        sc.nextLine();
        switch (a) {
            case 0:
                System.out.println(x:"Exiting");
                break;
            case 1:
                store.displayStore();
                int storechoice;
                do {
                    storeMenu();
                    storechoice = sc.nextInt();
                    sc.nextLine();
                    switch (storechoice) {
                        case 0:
                            break;
                        case 1:
                            System.out.print(s:"Enter the title of the media: ");
                            String title = sc.nextLine();

```

```

                System.out.print(s:"Enter the title of the media: ");
                String title = sc.nextLine();
                Media media = store.findMedia(title);
                if (media != null) {
                    System.out.println(media.toString());
                    int mediachoice;
                    do {
                        mediaDetailsMenu();
                        mediachoice = sc.nextInt();
                        sc.nextLine();
                        switch (mediachoice) {
                            case 1:
                                cart.addMedia(media);
                                break;
                            case 2:
                                if (media instanceof Playable) {
                                    ((Playable) media).play();
                                } else {
                                    System.out.println(x:"This media cannot be played");
                                }
                                break;
                            case 0:
                                break;
                            default:
                                System.out.println(x:"Invalid choice.");
                                break;
                        }
                    } while (mediachoice != 0);
                } else {
                    System.out.println(x:"Media not found");
                }
            } break;
            case 2:
                System.out.println(x:"Enter the title of the media: ");
                String title1 = sc.nextLine();
                Media media1 = store.findMedia(title1);
                if (media1 != null) {
                    System.out.println(media1.toString());
                    cart.addMedia(media1);

```

```
public class Aims {
    public static void main(String[] args) {
        case 2:
            System.out.println("Enter the title of the media: ");
            String title1 = sc.nextLine();
            Media media1 = store.findMedia(title1);
            if (media1 != null) {
                System.out.println(media1.toString());
                cart.addMedia(media1);
            } else {
                System.out.println("Media not found");
            }
            break;
        case 3:
            System.out.println("Enter the title of the media");
            String title2 = sc.nextLine();
            Media media2 = store.findMedia(title2);
            if (media2 != null) {
                if (media2 instanceof Playable) {
                    ((Playable) media2).play();
                } else {
                    System.out.println("This media cannot be played");
                }
            } else {
                System.out.println("Media not found");
            }
            break;
        case 4:
            cart.displayCart();
            int cartchoice;
            do {
                cartMenu();
                cartchoice = sc.nextInt();
                sc.nextLine();
                switch (cartchoice) {
                    case 0:
                        break;
                    case 1:
                        cart.filterCart(sc);
                        break;
                    case 2:
                        cart.sortCart(sc);

```

```

                } while (cartchoice != 0);
            }
            default:
                System.out.println("Invalid choice.");
        }
    }
}
```

```
public class Aims {
    public static void main(String[] args) {
        case 2:
            System.out.println(x:"1. Add media");
            System.out.println(x:"2. Remove media");
            System.out.println(x:"3. Back");
            int updatechoice = sc.nextInt();
            sc.nextLine();
            switch (updatechoice) {
                case 1:
                    store.addMedia(sc);
                    break;
                case 2:
                    store.removeMedia(sc);
                    break;
                case 3:
                    break;
                default:
                    System.out.println(x:"Invalid choice");
                    break;
            }
            break;
        case 3:
            cart.displayCart();
            cartMenu();
            int cartchoice = sc.nextInt();
            sc.nextLine();
            switch (cartchoice) {
                case 1:
                    cart.filterCart(sc);
                    break;
                case 2:
                    cart.sortCart(sc);
                    break;
                case 3:
                    cart.removeCart(sc);
                    break;
                case 4:
                    cart.playMedia(sc);
                    break;
                case 5:
                    if (cart.isEmpty()) {
```

```
                    }
                    if (cart.isEmpty()) {
                        System.out.println(x:"No item in cart");
                    } else {
                        System.out.println(x:"An order is created");
                        cart.emptyCart();
                    }
                    break;
                default:
                    break;
            }
            break;
        default:
            System.out.println(x:"Invalid choice.");
            break;
    }
} while (a != 0);
sc.close();
}
```

## 1.2 Cart class

```
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AIMS Project > src > hust > soict > dsai > aims > cart > J Cart.java > Cart > placeOrder(Scanner)
package hust.soict.dsai.aims.cart;

import hust.soict.dsai.aims.media.Media;
import hust.soict.dsai.aims.media.MediaComparatorByCostTitle;
import hust.soict.dsai.aims.media.MediaComparatorByTitleCost;
import hust.soict.dsai.aims.media.Playable;

import java.util.Collections;
import java.util.ArrayList;
import java.util.Comparator;
import java.util.Scanner;

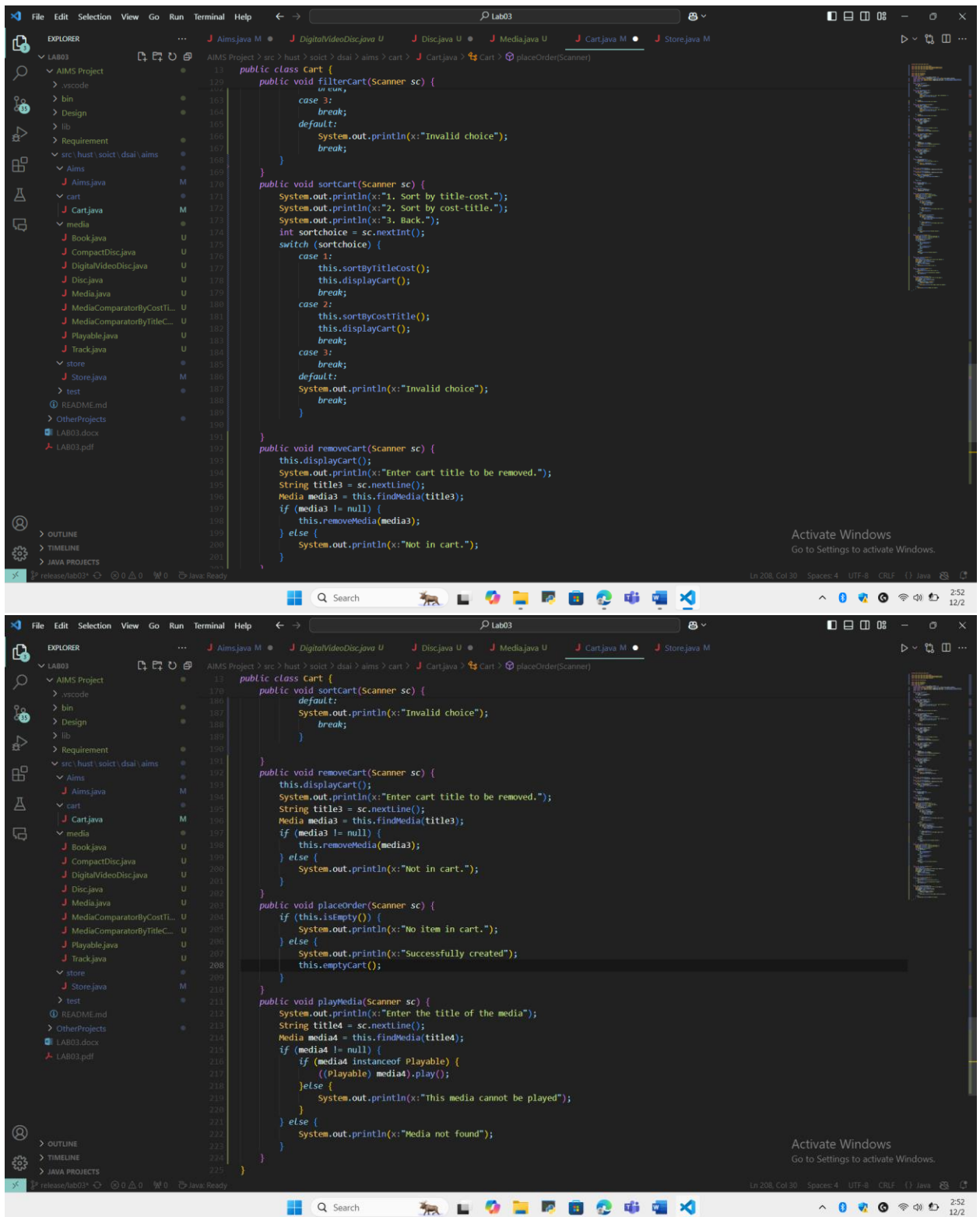
public class Cart {
    private static final int max_numbers_ordered = 20;
    private ArrayList<Media> itemsOrdered = new ArrayList<Media>(max_numbers_ordered);
    public static final Comparator<Media> Compare_by_title_cost = new MediaComparatorByTitleCost();
    public static final Comparator<Media> Compare_by_cost_title = new MediaComparatorByCostTitle();
    public Cart() {
    }
    public float totalCost() {
        float res = 0;
        for (Media m: itemsOrdered) {
            res += m.getCost();
        }
        return res;
    }
    public void searchMedia(int id) {
        boolean found = false;
        for (Media m: itemsOrdered) {
            if (m.getId() == id) {
                found = true;
                System.out.println("Disc in cart. Disc information: ");
                System.out.println(m.toString());
                break;
            }
        }
        if (!found) {
            System.out.println("Disc not found");
        }
    }
    public void searchMedia(String title) {
        boolean found = false;
    }
}
```

```
File Edit Selection View Go Run Terminal Help Lab03
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AIMS Project > src > hust > soict > dsai > aims > cart > J Cart.java > Cart > placeOrder(Scanner)
public class Cart {
    public void searchMedia(String title) {
        boolean found = false;
        for (Media m: itemsOrdered) {
            if (m.getTitle().equalsIgnoreCase(title)) {
                found = true;
                System.out.println("Disc in cart. Disc information: ");
                System.out.println(m.toString());
                break;
            }
        }
        if (!found) {
            System.out.println("Disc not found");
        }
    }
    public void addMedia(Media media) {
        boolean found = false;
        for (Media m: itemsOrdered) {
            if (m.equals(media)) {
                found = true;
                break;
            }
        }
        if (found) {
            System.out.println("Already in cart");
        } else {
            itemsOrdered.add(media);
            System.out.println("Added successfully");
        }
    }
    public void removeMedia(Media media) {
        boolean found = false;
        for (Media m: itemsOrdered) {
            if (m.equals(media)) {
                found = true;
                break;
            }
        }
        if (found) {
            itemsOrdered.remove(media);
            System.out.println("Removed successfully");
        } else {
        }
    }
}
```

```
public class Cart {  
    public void removeMedia(Media media) {  
        System.out.println(x:"Removed successfully");  
    } else {  
        System.out.println(x:"Not in cart");  
    }  
}  
  
public boolean Contains(Object o) {  
    boolean found = false;  
    for (Media m: itemsOrdered) {  
        if (m.equals(o)) {  
            found = true;  
            break;  
        }  
    }  
    return found;  
}  
  
public void displayCart() {  
    if (itemsOrdered.size() == 0) {  
        System.out.println(x:"No item in cart.");  
    } else {  
        for (Media m: itemsOrdered) {  
            System.out.println(m.toString());  
        }  
    }  
}  
  
public void sortByTitleCost() {  
    Collections.sort(itemsOrdered, Compare_by_title_cost);  
}  
  
public void sortByCostTitle() {  
    Collections.sort(itemsOrdered, Compare_by_cost_title);  
}  
  
public Media findMedia(String title) {  
    for (Media m: itemsOrdered) {  
        if (m.getTitle().equalsIgnoreCase(title)) {  
            return m;  
        }  
    }  
    return null;  
}  
  
public boolean isEmpty() {  
    if (itemsOrdered.size() == 0) {  
        return true;  
    }  
    return false;  
}
```

```
public class Cart {  
    public boolean isEmpty() {  
        if (itemsOrdered.size() == 0) {  
            return true;  
        }  
        return false;  
    }  
  
    public void emptyCart() {  
        for (Media m: itemsOrdered) {  
            itemsOrdered.remove(m);  
        }  
    }  
  
    public void filterCart(Scanner sc) {  
        System.out.println(x:"1. Filter by id");  
        System.out.println(x:"2. Filter by title");  
        System.out.println(x:"3. Back");  
        int filterchoice = sc.nextInt();  
        switch (filterchoice) {  
            case 1:  
                int id = sc.nextInt();  
                Cart cart = new Cart();  
                for (Media m: itemsOrdered) {  
                    if (m.getId() == id) {  
                        cart.addMedia(m);  
                    }  
                }  
                if (cart.isEmpty()) {  
                    System.out.println(x:"No media same title");  
                } else {  
                    cart.displayCart();  
                }  
                break;  
            case 2:  
                String title = sc.nextLine();  
                Cart cart1 = new Cart();  
                for (Media m: itemsOrdered) {  
                    if (m.getTitle().equalsIgnoreCase(title)) {  
                        cart1.addMedia(m);  
                    }  
                }  
                if (cart1.isEmpty()) {  
                    System.out.println(x:"No media same title");  
                } else {  
                    cart1.displayCart();  
                }  
                break;  
            case 3:  
                return;  
            default:  
                System.out.println(x:"Invalid choice");  
        }  
    }  
}
```



### 1.3 Store class

```
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EXPLORER AIMS Project > src > hust > soict > dsai > aims > store > Store.java > Store > addMedia(Scanner)
  AIMS Project
  > .vscode
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    J Aims.java
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    J Cart.java
    > media
    J Book.java
    J CompactDisc.java
    J DigitalVideoDisc.java
    J Disc.java
    J Media.java
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    J MediaComparatorByTitleC...
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  > JAVA PROJECTS
  release/Lab03*
  0 0 0 0 0 Java Ready

package hust.soict.dsai.aims.store;
import java.util.ArrayList;
import java.util.Scanner;

import hust.soict.dsai.aims.media.Book;
import hust.soict.dsai.aims.media.CompactDisc;
import hust.soict.dsai.aims.media.DigitalVideoDisc;
import hust.soict.dsai.aims.media.Media;

public class Store {
    private ArrayList<Media> itemsInStore;
    public Store() {
        itemsInStore = new ArrayList<>();
    }
    public void addMedia(Media media) {
        boolean found = false;
        for (Media m : itemsInStore) {
            if (m.equals(media)) {
                found = true;
                break;
            }
        }
        if (found) {
            System.out.println(x:"The dvd is already added");
        } else {
            itemsInStore.add(media);
            System.out.println(x:"The dvd has been added");
        }
    }
    public void removeMedia(Media media) {
        boolean found = false;
        for (Media m : itemsInStore) {
            if (m.equals(media)) {
                found = true;
                break;
            }
        }
        if (found) {
            itemsInStore.remove(media);
            System.out.println(x:"The dvd has been removed");
        } else {
            System.out.println(x:"DVD not found");
        }
    }
}
```

Activate Windows  
Go to Settings to activate Windows.

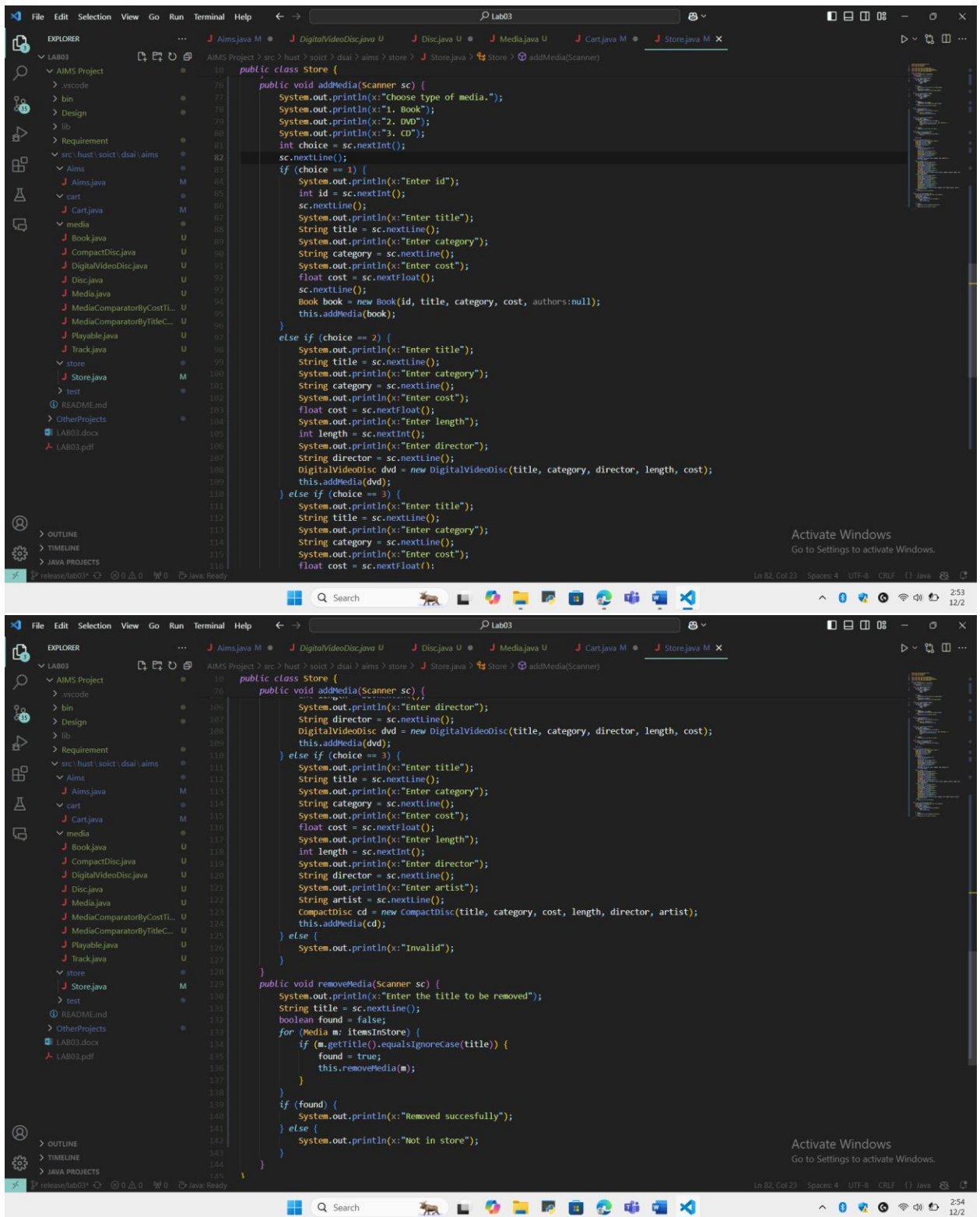
Ln 82, Col 23 Spaces: 4 UTF-8 CRLF (1) Java 253

```
File Edit Selection View Go Run Terminal Help Lab03
EXPLORER AIMS Project > src > hust > soict > dsai > aims > store > Store.java > Store > addMedia(Scanner)
  AIMS Project
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    Aims
    J Aims.java
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    J Cart.java
    > media
    J Book.java
    J CompactDisc.java
    J DigitalVideoDisc.java
    J Disc.java
    J Media.java
    J MediaComparatorByCostTL...
    J MediaComparatorByTitleC...
    J Playable.java
    J Track.java
    > store
    J Store.java
    > test
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    > OtherProjects
    LAB03.docx
    LAB03.pdf
  > OUTLINE
  > TIMELINE
  > JAVA PROJECTS
  release/Lab03*
  0 0 0 0 0 Java Ready

public class Store {
    public void displayStore() {
        if (itemsInStore.size() == 0) {
            System.out.println(x:"No item in store.");
        } else {
            for (Media m : itemsInStore) {
                System.out.println(m.toString());
            }
        }
    }
    public void searchMedia(String title) {
        boolean found = false;
        for (Media m : itemsInStore) {
            if (m.getTitle().equalsIgnoreCase(title)) {
                found = true;
                System.out.println(x:"Disc in cart. Disc information: ");
                System.out.println(m.toString());
                break;
            }
        }
        if (!found) {
            System.out.println(x:"Disc not found");
        }
    }
    public Media findMedia(String title) {
        for (Media m : itemsInStore) {
            if (m.getTitle().equalsIgnoreCase(title)) {
                return m;
            }
        }
        return null;
    }
    public void addMedia(Scanner sc) {
        System.out.println(x:"Choose type of media.");
        System.out.println(x:"1. Book");
        System.out.println(x:"2. DVD");
        System.out.println(x:"3. CD");
        int choice = sc.nextInt();
        sc.nextLine();
        if (choice == 1) {
            System.out.println(x:"Enter id");
        }
    }
}
```

Activate Windows  
Go to Settings to activate Windows.

Ln 82, Col 23 Spaces: 4 UTF-8 CRLF (1) Java 253



## 1.4 Media class

```
package hust.soiict.dsai.aims.media;

public abstract class Media {
    private int id;
    private String title;
    private String category;
    private float cost;
    public Media(int id, String title, String category, float cost) {
        this.id = id;
        this.title = title;
        this.category = category;
        this.cost = cost;
    }
    public int getId() { ... }
    public void setId(int id) { ... }
    public String getTitle() { ... }
    public void setTitle(String title) { ... }
    public String getCategory() { ... }
    public void setCategory(String category) { ... }
    public float getCost() { ... }
    public void setCost(float cost) {
        this.cost = cost;
    }
    public String toString() {
        return "id: " + id + "\n"
            + "title: " + title + "\n"
            + "category: " + category + "\n"
            + "cost: " + cost + ".";
    }
    public boolean equals(Object o) {
        if (this == o) {
            return true;
        }
        if (o == null || this.getClass() != o.getClass()) {
            return false;
        }
        Media media = (Media) o;
        return title.equals(media.getTitle());
    }
}
```

## 1.5 Disc class

```
package hust.soiict.dsai.aims.media;

public class Disc extends Media {
    private int length;
    private String director;
    public int getLength() {
        return length;
    }
    public String getDirector() {
        return director;
    }
    public Disc(int Id, String title, String category, float cost, int length, String director) {
        super(Id, title, category, cost);
        this.length = length;
        this.director = director;
    }
    public void play() {
        System.out.println("Playing Disc: " + this.getTitle());
        System.out.println("Disc length: " + this.getLength());
    }
    public String toString() {
        return "id: " + this.getId() + "\n"
            + "title: " + this.getTitle() + "\n"
            + "category: " + this.getCategory() + "\n"
            + "cost: " + this.getCost() + "\n"
            + "length" + this.getLength() + "\n"
            + "Director" + this.getDirector() + ".";
    }
}
```

## 1.6 Dvd class

```
package hust.soict.dsai.aims.media;

public class DigitalVideoDisc extends Disc implements Playable {
    private static int nbDigitalVideoDisc;
    public DigitalVideoDisc(String title) {
        nbDigitalVideoDisc++;
        super(nbDigitalVideoDisc, title, category:null, cost:0, length:0, director:null);
    }

    public DigitalVideoDisc(String category, String title, float cost) {
        nbDigitalVideoDisc++;
        super(nbDigitalVideoDisc, title, category, cost, length:0, director:null);
    }

    public DigitalVideoDisc(String director, String category, String title, float cost) {
        nbDigitalVideoDisc++;
        super(nbDigitalVideoDisc, title, category, cost, length:0, director);
    }

    public DigitalVideoDisc(String title, String category, String director, int length, float cost) {
        nbDigitalVideoDisc++;
        super(nbDigitalVideoDisc, title, category, cost, length, director);
    }

    public String toString() {
        return "Id: " + this.getId() + "\n"
            + "Title: " + this.getTitle() + "\n"
            + "Category: " + this.getCategory() + "\n"
            + "Cost: " + this.getCost() + "\n"
            + "Length: " + this.getLength() + "\n"
            + "Director: " + this.getDirector() + ".";
    }

    public boolean isMatch(String title) {
        if (title.equals(this.getTitle())) {
            return true;
        } else {
            return false;
        }
    }

    public void play() {
        System.out.println("Playing DVD: " + this.getTitle());
        System.out.println("DVD length: " + this.getLength());
    }
}
```

## 1.7 CD class

```
package hust.soict.dsai.aims.media;
import java.util.ArrayList;

public class CompactDisc extends Disc implements Playable {
    private static int nbDisc;
    private String artist;
    private ArrayList<Track> tracks;
    public CompactDisc(String title, String category, float cost, int length, String director, String artist) {
        nbDisc++;
        this.artist = artist;
        super(nbDisc, title, category, cost, length, director);
    }

    public String getArtist() {
        return artist;
    }

    public void addTrack(Track track) {
        boolean found = false;
        for (Track t: tracks) {
            if (t.equals(track)) {
                found = true;
                break;
            }
        }
        if (found) {
            System.out.println(x:"Already in track");
        } else {
            tracks.add(track);
            System.out.println(x:"Added successfully");
        }
    }

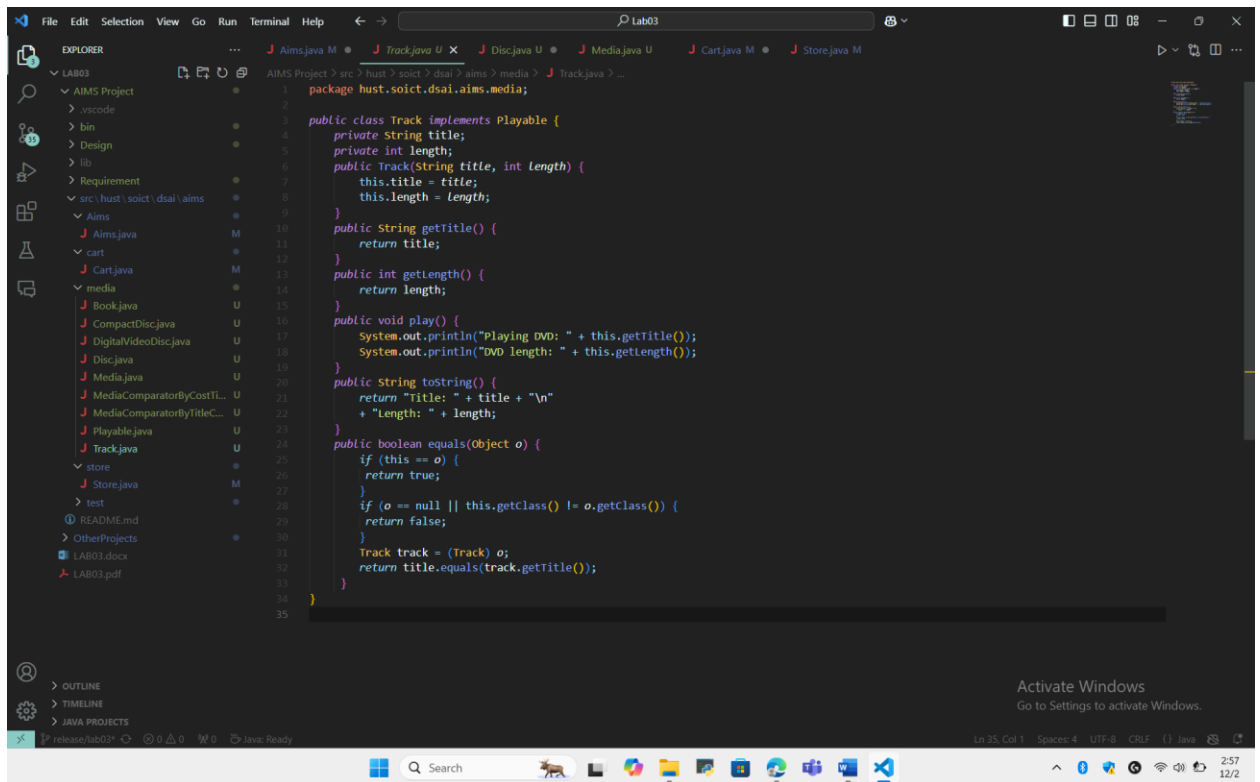
    public void removeTrack(Track track) {
        boolean found = false;
        for (Track t: tracks) {
            if (t.equals(track)) {
                found = true;
                break;
            }
        }
        if (found) {
            tracks.remove(track);
            System.out.println(x:"Removed successfully");
        } else {
            System.out.println(x:"Not in track");
        }
    }
}
```

```
public class CompactDisc extends Disc implements Playable {  
    public void removeTrack(Track track) {  
        // ...  
    }  
    public int getLength() {  
        int length = 0;  
        for (Track t: tracks) {  
            length += t.getLength();  
        }  
        return length;  
    }  
    public void play() {  
        for (Track t: tracks) {  
            t.play();  
        }  
    }  
    public String toString() {  
        StringBuilder sb = new StringBuilder();  
        sb.append(str:"Id: ").append(nbDisc).append(str:"\n")  
        .append(str:"Title: ").append(this.getTitle()).append(str:"\n")  
        .append(str:"Category: ").append(this.getCategory()).append(str:"\n")  
        .append(str:"Cost: ").append(this.getCost()).append(str:"\n")  
        .append(str:"Length: ").append(this.getLength()).append(str:"\n")  
        .append(str:"Director: ").append(this.getDirector()).append(str:"\n")  
        .append(str:"Artist: ").append(this.artist).append(str:"\n");  
        for (Track t: tracks) {  
            sb.append(t.toString());  
        }  
        return sb.toString();  
    }  
    public boolean contains(Object o) {  
        boolean found = false;  
        for (Track t: tracks) {  
            if (t.equals(o)) {  
                found = true;  
                break;  
            }  
        }  
        return found;  
    }  
}
```

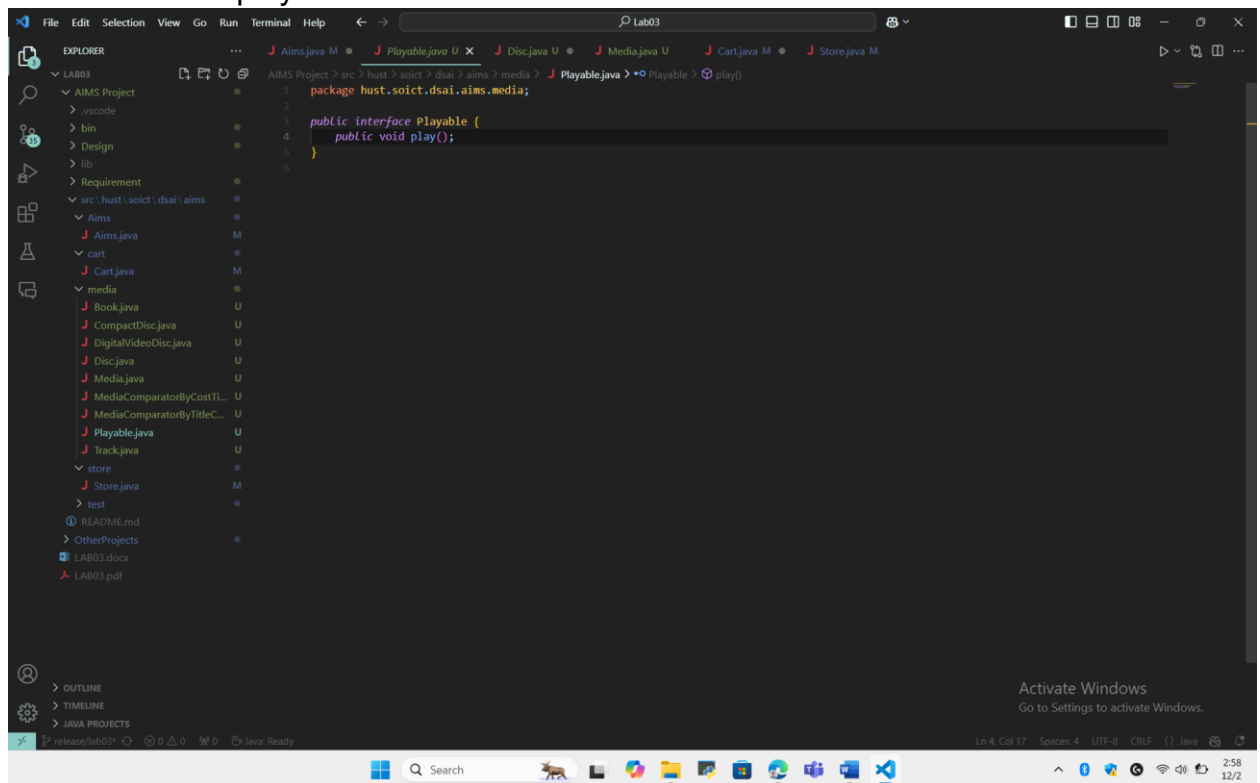
## 1.8 Book class

```
public class Book extends Media {  
    private List<String> authors = new ArrayList<String>();  
    public Book(int id, String title, String category, float cost, List<String> authors) {  
        super(id, title, category, cost);  
        this.authors = authors;  
    }  
    public void addAuthor(String authorName) {  
        boolean found = false;  
        for (String name : authors) {  
            if (name.equals(authorName)) {  
                found = true;  
                break;  
            }  
        }  
        if (found) {  
            System.out.println(x:"Author already in list");  
        } else {  
            authors.add(authorName);  
            System.out.println(x:"Add successfully");  
        }  
    }  
    public void removeAuthor(String authorName) {  
        boolean found = false;  
        for (String name : authors) {  
            if (name.equals(authorName)) {  
                found = true;  
                break;  
            }  
        }  
        if (found) {  
            authors.remove(authorName);  
            System.out.println(x:"Remove successfully");  
        } else {  
            System.out.println(x:"Author not in list");  
        }  
    }  
    public List<String> getAuthors() {  
        return authors;  
    }  
    public void setAuthors(List<String> authors) {  
        this.authors = authors;  
    }  
}
```

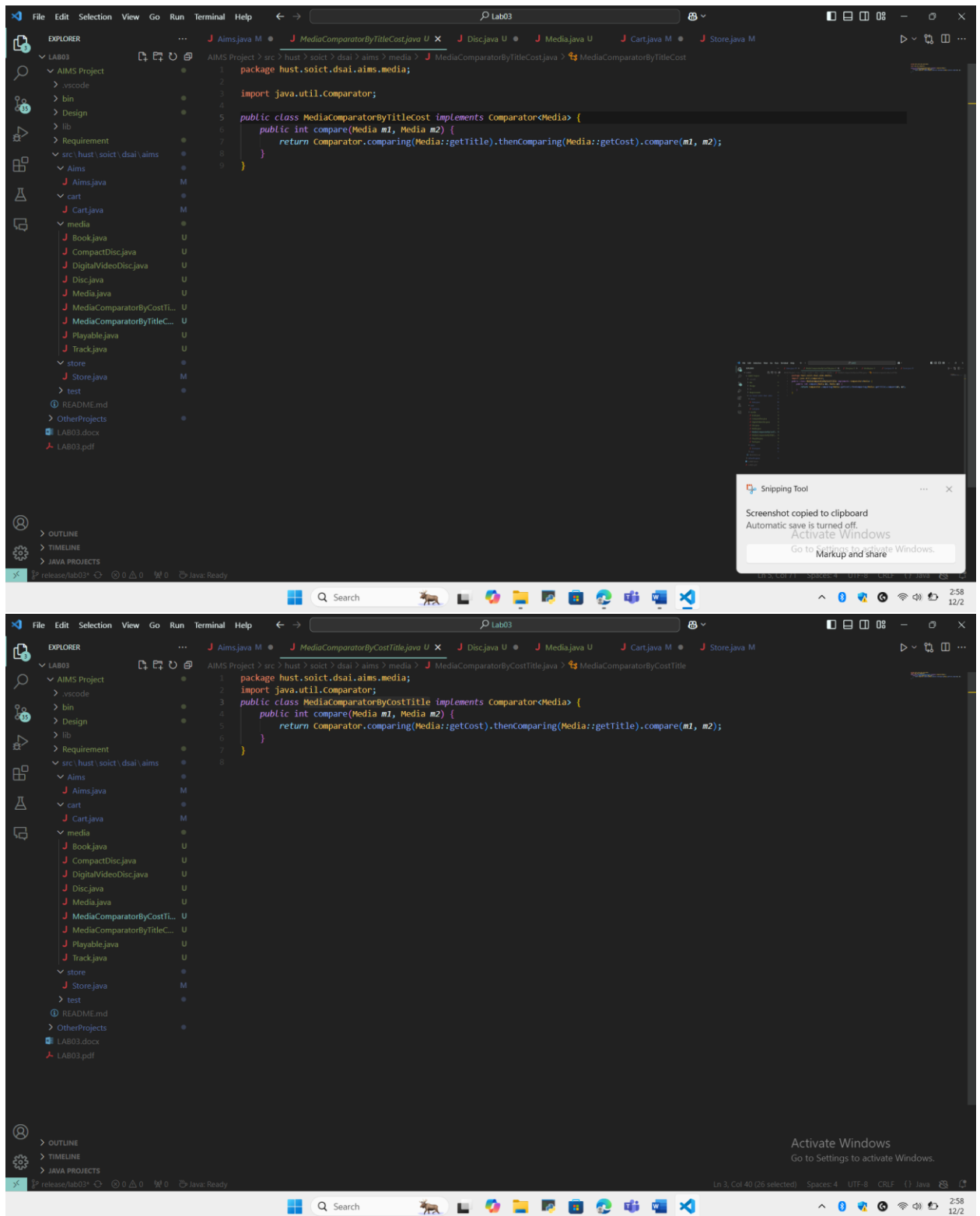
## 1.9 Track class



## 1.10 Interface playable

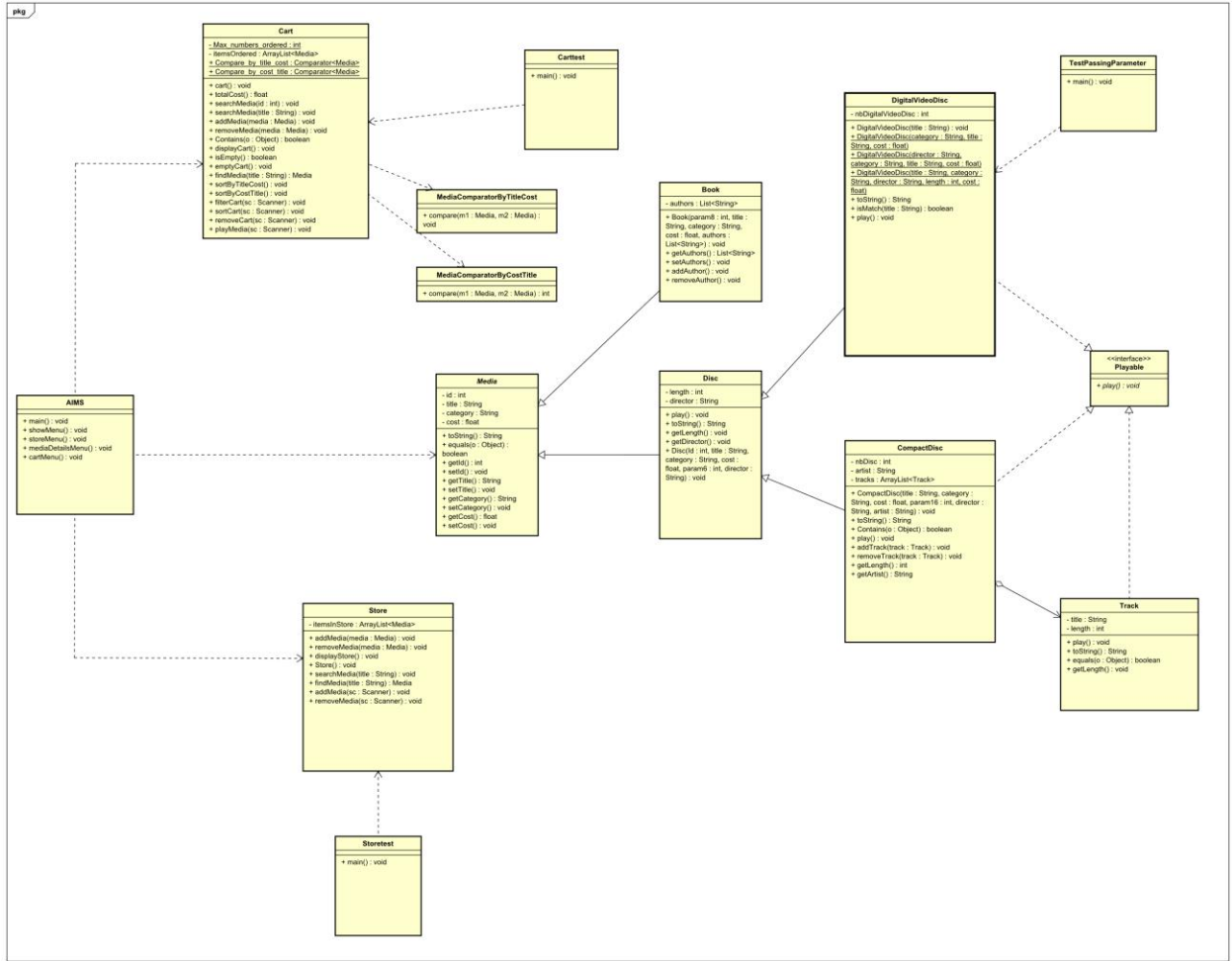


## 1.11 Comparator

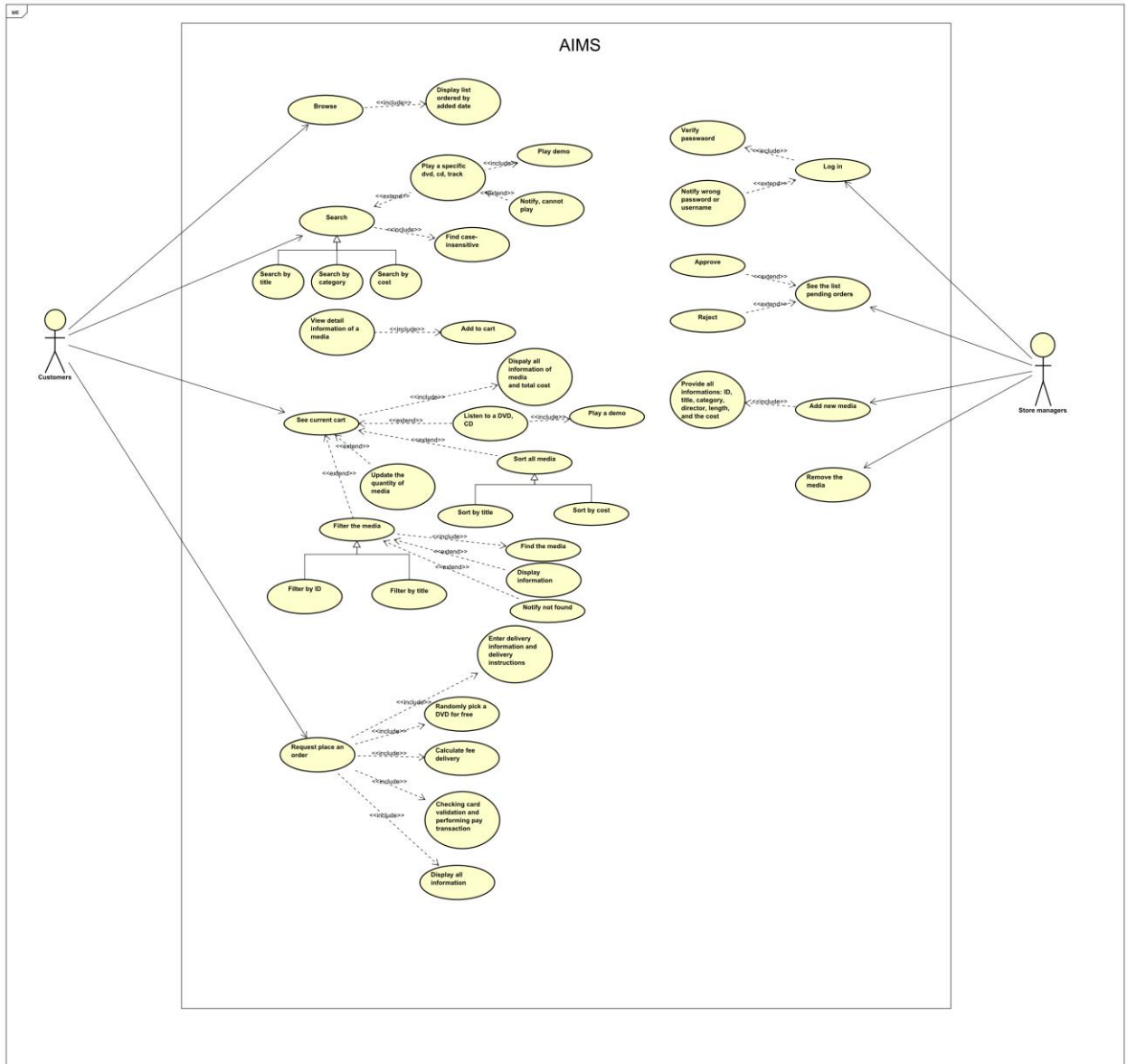


## 2. Diagram

### 2.1 Class diagram



## 2.2 Usecase diagram



3. Answer

3.1 Which classes are aggregates of other classes? Checking all constructors of whole classes if they initialize for their parts?

- Cart aggregates Media objects
- Store aggregates Media objects
- CompactDisc aggregates Track objects

3.2 When overriding the equals() method of the Object class, you will have to cast the Object parameter obj to the type of Object that you are dealing with. For example, in the Media class, you must cast the Object obj to a Media, and then check the equality of the two objects' attributes as the above requirements (i.e. title for Media; title and length for Track). If the passing object is not an instance of Media, what happens?

- If the passing object is not an instance of Media, it will return false.

3.3 **Question:** Alternatively, to compare items in the cart, instead of using Comparator, we can use the Comparable interface and override the compareTo() method. You can refer to the Java docs to see the information of this interface.

Suppose we are taking this Comparable interface approach.

- What class should implement the Comparable interface?
- In those classes, how should you implement the compareTo() method to reflect the ordering that we want?
- Can we have two ordering rules of the item (by title then cost and by cost then title) if we use this Comparable interface approach?
- Suppose the DVDs has a different ordering rule from the other media types, that is by title, then decreasing length, then cost. How would you modify your code to allow this?

- Class media should implement the Comparable interface and subclass of media class such as Book, Disc, DVD, CD should also override that method.

- Each class should override the compareTo() method to define the natural ordering, for example by title, by cost, ...

- Using Comparable interface, a class can define only one ordering using compareTo() method.

- We should override this method for DVD class

```
public int compareTo(Media media) {  
    if (media instanceof DigitalVideoDisc) {  
        DigitalVideoDisc dvd = (DigitalVideoDisc) media;  
        int titleCompare = this.getTitle().compareTo(dvd.getTitle());  
        if (titleCompare != 0) {  
            return titleCompare;  
        }  
        int lengthCompare = Integer.compare(dvd.getLength(), this.getLength());  
        if (lengthCompare != 0) {  
            return lengthCompare;  
        }  
        return Float.compare(this.getCost(), dvd.getCost());  
    }  
    return super.compareTo(media);  
}
```